ANNA SHEAFFER

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EDUCATION

Tufts University | Medford, MA
MS, Computer Science, May 2026
PhD, Computer Science, May 2029

Illinois Wesleyan University | Bloomington, IL — Cumulative GPA: **3.78**/4.0
BS, Computer Science and Psychology, May 2024 — Dean's List Fall 2020 - Spring 2024

Relevant Coursework: Deep Learning, Database Systems, Computer Organization-Architecture, Software Development, Algorithm Design and Analysis, Behavioral Neuroscience, Medical Neuroscience

RESEARCH AND TEACHING EXPERIENCE

Graduate Researcher — August 2024 – *Present*
Department of Computer Science, Tufts University, Medford, MA

- Engineer a real-time multimodal sensor acquisition framework in Python
- Leverage multithreading and Lab Streaming Layer (LSL) to synchronize and manage concurrent data streams across heterogeneous physiological sensors
- Identify hemodynamic biomarkers from fNIRS data using HOMER3 and Matlab
- Perform fNIRS channel selection using statistically significant oxygenation changes to inform feature selection for downstream machine learning classification models
- Design and execute IRB-compliant human subjects research protocols, ensuring ethical data collection and adherence to institutional research standards

Graduate Teaching Assistant — August 2024 – *Present*
Department of Computer Science, Tufts University, Medford, MA

- Lead weekly discussion sessions for a seminar course, reinforcing core course concepts
- Evaluate assignments, exams, and projects, providing detailed feedback to support academic growth
- Develop and refine rubrics to ensure consistent and equitable grading
- Respond to student inquiries and resolve course-related issues electronically and in office hours, maintaining a quick turnaround on communications
- Maintain accurate grade records using Canvas

Summer Research Intern — June 2026 – August 2026
Center for Aphasia Research and Treatment, Shirley Ryan AbilityLab, Chicago, IL

- Designed and built a cross-platform desktop application in Python (PyQt) that integrates multiple text-to-speech engines, giving users a choice of voices, images and speech output.
- Partnered with speech-language pathologists and people with aphasia in iterative co-design sessions, translating their feedback into features that served both clinical goals and communication needs
- Applied participatory design methods to build accessible software for users with speech-related disabilities

Independent Research — May 2022 – May 2024
Department of Computer Science, Illinois Wesleyan University, Bloomington, IL

- Engineer data processing workflows in Python using pandas and NumPy, encompassing data cleaning, missing value imputation, cross-database integration, and transformation of biological datasets into frontend-ready formats
- Develop frontend using Javascript, HTML, and CSS that displays visualizations of data analysis that enable researchers to intuitively explore and interpret large-scale datasets
- Implement interactive biological network graph and alignment visualizations to communicate novel and complex relational data structures to support biology research

Publications

[1] A. Sheaffer, A. Kim, M. Brower, M. Russell, T. Brunyé, and R. J. K. Jacob, "Assessing cognitive workload during dual-task performance: An fNIRS study of N-back tasks under walking and standing conditions," in *Proc. Frontiers in Neuroergonomics*, 2026, to be published.

Honors and Awards

NexSTEM Scholarship Recipient | *National Science Foundation* August 2022 – May 2024

Alumni Scholarship | *Illinois Wesleyan University, Bloomington, IL* August 2020 – May 2024

Eckley Summer Scholar | *Illinois Wesleyan University, Bloomington, IL* May 2023 - August 2023

ACM-BCB Undergraduate Travel Grant | *Chicago, IL* August 2022

Barry Goldwater Scholarship Nominee | *Illinois Wesleyan University, Bloomington, IL* January 2022

Technical Skills

Python, Java, C++, Javascript, HTML/CSS, SQL, Matlab | Numpy, Pandas, Scikit-learn | Excel, Qualtrics, Git, HOMER3